Notes on Distributions, Sobolev Spaces and Weak Solutions

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The main objective of this text is to compile everything I've read or discussed about weak solutions on PDEs and the prerequisites for understanding this topic.

Summary

CHAPTER 1

Motivation

We will begin by studying the solution of the following equation:

$$u_t + uu_x = 0, \text{ for } x \in \mathbb{R} \text{ and } t > 0 \tag{1}$$

This is commonly known as the inviscid Burger's equation, a simplified model for fluid mechanics, modelling one dimensional flow where $u(x,t)$ is the velocity of the particle located in x at time t. So, first, we are going to try to find regular solutions for this equation by the method of characteristics.

Considering $u(x,t)$ to be a smooth solution for (1) satisfying the initial condition problem:

$$u(x,0) = u_0(x)$$

Where $u_0(x)$ is a function of x. We'll start by defining the characteristic curves:

$$\begin{cases} \dot{x}(t) = u(x(t),t),\ x(0) = x_0 \\ \dot{y}(t) = 1,\ y(0) = y_0 \end{cases}$$

Solely from that, we get $y = t$. And now, by calculating the total derivative of u on the curves just defined, we obtain:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\partial u}{\partial t}\frac{\partial y}{\partial t} + \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} = u_t + u_x u = 0$$

From this, two conclusions are made: u is constant along the curve $(x(t),t)$, which implies that $\dot{x}(t) = u_0(x_0)$. By solving this last ODE, we conclude that:

$$x = x_0 + u_0(x_0)t$$

And for all points in this curve $u(x,t) = u_0(x_0)$. For convenience's sake, we will choose this as our initial curve u_0:

Using the initial condition given by u_0, the characteristic curves are:

We can clearly see that there are curves intersecting each other, so the value of $u(x, t)$ is not well-defined at the intersection points, therefore this cannot be a solution for our equation.

Now, the question that rises is: "if that isn't a solution, then what is?". And that will motivate our next few chapters of functional analysis definitions and theorems, aiming to build a generalized perspective on what is a solution for a PDE.

CHAPTER 2

Test functions and Distributions

Our main objective in this chapter is to make useful definitions to generalize the derivative of a function. For that, we will need a mechanism that can weaken the conditions for a function to have partial derivatives, by removing it entirely and passing it to another convenient function that always has a derivative in any axis.

2.1 The elements of $\mathscr{D}(\Omega)$ and $\mathscr{D}'(\Omega)$

Definition 2.1: Test functions

Let Ω be a subset of $\mathbb{R}^n$. Then $\mathscr{D}(\Omega)$ is the set of all C^∞ functions defined on Ω with compact support.

An example of such function is:

$$f(x) = \begin{cases} \exp(\frac{-a^2}{a^2 - x^2}), |x| < a \\ 0, |x| \geq a \end{cases}$$

And, for any φ on this set and any locally integrable function $f : \mathbb{R}^n \to \mathbb{R}$, using integration by parts we get:

$$\int_K D^\alpha f \varphi = (-1)^{|\alpha|} \int_K f D^\alpha \varphi \tag{2}$$

Where $K \supseteq \text{supp}\varphi$ is a compact subset of Ω and α is a multi-index. This will motivate our next definition, but before that, we need to quickly describe the topology of $\mathscr{D}(\Omega)$:

A sequence $\{\varphi_n\}_{n \in \mathbb{N}}$ in $\mathscr{D}(\Omega)$ converges to φ if there exists a compact set $K \subset \Omega$ such that $\text{supp}(\varphi_n) \in K$ for all n and $D^\alpha \varphi_n \to D^\alpha \varphi$ for all α.

Now we can define the most important tool used in this endeavour:

Definition 2.2: Distributions

A linear functional T is said to be a distribution if for all sequences $\{\varphi_n\}_{n\in\mathbb{N}}$ in $\mathscr{D}(\Omega)$ convergent to some φ, the sequence $\{T\varphi_n\}_{n\in\mathbb{N}}$ converges to $T\varphi$ in $\mathscr{D}(\Omega)$.

We will use $\mathscr{D}'(\Omega)$ for the set of all distributions. Note that $\mathscr{D}'(\Omega)$ uses the same notation as the dual space of $\mathscr{D}(\Omega)$ and that is no coincidence, as the distributions are precisely the dual elements of the test functions, as they are the linear functionals defined from $\mathscr{D}(\Omega)$ to $\mathbb{R}$. Furthermore, it is worth to point that both $\mathscr{D}(\Omega)$ and $\mathscr{D}'(\Omega)$ are vector spaces over $\mathbb{R}$.

Now, aiming for (2), we will define the following:

Definition 2.3: Distribution given by a function

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a locally integrable function, we say that the distribution given by f is:

$$T_f(\varphi) = \int_\Omega f\varphi$$

It is trivial to verify that T_f is in fact a distribution, because as $\varphi_n \to \varphi$, $\int_\Omega f\varphi_n \to \int_\Omega f\varphi$.

Another useful distribution is the Dirac "δ-function", which is in fact a distribution given by:

Definition 2.4: The Dirac distribution

For any $\varphi \in \mathscr{D}(\Omega)$:

$$\delta(\varphi) = \varphi(0)$$

Both distributions will be essential for the next sections.

2.1.1 Multiplication of a distribution by a function

From now on, we will use the notation $\mathscr{E}(\Omega)$ for the C^∞ functions defined in Ω. First, let $\{\varphi_n\}_{n\in\mathbb{N}}$ be a sequence of functions in $\mathscr{D}(\Omega)$ converging to φ, notice that, for $\psi \in \mathscr{E}(\Omega)$, $\{\psi\varphi_n\}_{n\in\mathbb{N}}$ converges to φ. Additionally, as φ_n has compact support, $\psi\varphi$ will also. Therefore, if T is a distribution, the sequence $\{T(\psi\varphi_n)\}_{n\in\mathbb{N}}$ is convergent to $T(\psi\varphi)$. So:

$$\psi T(\varphi) = T(\psi\varphi)$$

defines a distribution. If T is a distribution given by a function f, we get that:

$$\psi T_f(\varphi) = T_f(\psi\varphi) = \int_\Omega f\psi\varphi = T_{f\psi}(\varphi)$$

2.2 Distributional derivatives

Having (2) in mind, let's build a definition of a distribution derivative that can satisfy that equation. Let f be a smooth function in Ω, so:

$$D^\alpha Tf(\varphi) = D^\alpha \int_\Omega f\varphi$$

Now, by the dominated convergence theorem we can push the derivative inside the integral, that is:

$$D^\alpha \int_\Omega f\varphi = \int_\Omega D^\alpha f\varphi = (-1)^{|\alpha|} \int_K f D^\alpha \varphi$$

So, we now know what we have to define $D^\alpha T$ as to serve our needs. We let f as a smooth function just so $D^\alpha f$ has some meaning in a sense, but in general, the only condition f must abide is to be locally integrable.

Definition 2.5: Derivative of a distribution

Let $T \in \mathscr{D}'(\Omega)$, the α derivative of T is:

$$D^\alpha T(\varphi) = (-1)^{|\alpha|} T(D^\alpha \varphi)$$

Another important thing to note is that, for a distribution given by a function f, the following identity is verified:

Lemma 2.1

$$D^\alpha T_f(\varphi) = T_{D^\alpha f} = (-1) T_f(D^\alpha \varphi)$$

Which was already proven before. In addition to that, we will say that a locally integrable function f has a distributional derivative if there exists a distribution G such that for all $\varphi \in \mathscr{D}(\Omega)$:

$$\frac{\mathrm{d}}{\mathrm{d}t} T_f(\varphi) = G(\varphi)$$

So G is the distributional derivative of f. If $G = T_g$ for some locally integrable function g, then we also say that g is the distributional derivative of f. This distributional derivative is also what is called the weak derivative of f, sometimes denoted as $\partial_t f$.

It should be clear that if a function has a regular derivative, its weak derivative will be the same, as the identity above will be satisfied by:

$$\frac{\mathrm{d}}{\mathrm{d}t} T_f = T_{\frac{\mathrm{d}f}{\mathrm{d}t}}$$

Example 2.1: The derivative of the Heaviside Function

The Heaviside function is defined as:

$$H(x) \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases}$$

So the distributional derivative of H is the δ distribuition.

Proof. Let $\Omega \in \mathbb{R}$. As H is a locally integrable function, let's evaluate the distribution given by H:

$$T_H(\varphi) = \int_\Omega H(x)\varphi(x)dx$$

So, by taking the derivative of this distribution and using lemma 2.1 we get:

$$\frac{\mathrm{d}}{\mathrm{d}x}T_H(\varphi) = -T_H(\frac{\mathrm{d}\varphi}{\mathrm{d}x}) = -\int_0^{+\infty} H(x)\frac{\mathrm{d}\varphi}{\mathrm{d}x}dx = \varphi(x)\Big|_0^{+\infty} = \varphi(0) = \delta(\varphi)$$

Therefore, δ is the distributional derivative of the Heaviside function.

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Example 2.2: The derivatives of the δ distribution

For $\Omega \subseteq \mathbb{R}$ and $\varphi \in \mathscr{D}(\Omega)$, using definition 2.5, we get:

$$\delta^{(n)}(\varphi(x)) = (-1)^n\delta(\varphi^{(n)}(x)) = (-1)^n\varphi^{(n)}(x)$$

Where $\varphi^{(n)}$ is the n-nht derivative of φ.

Proposition 2.1: The derivative of a distribuition multiplied by a $\mathscr{E}(\Omega)$ function

By definition 2.5:

$$D^\alpha(\psi T)(\varphi) = D^\alpha T(\psi\varphi) = (-1)^{|\alpha|}T(D^\alpha\psi\varphi) = (-1)^{|\alpha|}(D^\alpha\psi)T(\varphi)$$